National Institute of Neurological Disorders and Stroke

Multiple Sclerosis (MS)

Table of Contents

What is multiple sclerosis (MS)? 2

Who is more likely to get multiple sclerosis?.....6

How is multiple sclerosis diagnosed and treated? 8

What are the latest updates on multiple sclerosis? 16

How can I or my loved one help improve care
for people with multiple sclerosis? 20

Where can I find more information about multiple
multiple sclerosis? 21

What is multiple sclerosis (MS)?

Multiple sclerosis (MS) is a chronic neurological disorder. It's an autoimmune disorder, meaning that in MS, the immune system—which normally protects us from viruses, bacteria, and other threats—mistakenly attacks healthy cells. MS symptoms usually begin in young adults, between ages 20 and 40.

MS affects people differently. A small number of people with MS will have mild symptoms with little disability, but others will experience worsening symptoms that will lead to increased disability over time. Most people with MS have short periods of symptoms that resolve fully or partially after they appear. These periods are followed by long stretches without noticeable symptoms. Most people with MS have a normal life expectancy.

Myelin and the immune system

In MS, the immune system attacks myelin in the central nervous system. Myelin is a mixture of protein and fatty acids that makes up the protective cover (known as the myelin sheath) that coats nerve fibers (axons). Myelin is what gives the brain's white matter its whitish appearance and helps with communication between neurons. The central nervous system is made up of the brain and the spinal cord.

In addition to causing damage to the myelin sheath, MS also damages the nerve cell bodies, which are found in the brain's gray matter, as well as

the axons themselves. As the disease progresses, the outermost layer of the brain, called the cerebral cortex, shrinks. This process is known as cortical atrophy. The way that cortical atrophy happens in MS may connect it with some neurodegenerative disorders.

Sclerosis is a medical term for the distinctive areas of scar-like tissue (also called plaques or lesions) that result from the attack on myelin by the immune system. These areas are visible on an MRI (magnetic resonance imaging). The patches of scar-like tissue can be as small as the head of a pin or as large as a golf ball.

The symptoms of MS depend on the severity of the attacks as well as the location and size of the plaques.

Types of multiple sclerosis

The course of MS is different for each person, which makes it difficult to predict how a person will do with the disease. While many different courses or progressions of MS have been used over the years, these are changing as the scientific and medical community better understands different ways the disease can progress.

Currently, the five courses used to describe MS are:

1. **Clinically isolated syndrome:** Symptoms come from a single attack (also called "exacerbation" or "relapse") followed by complete or near-complete recovery. MRI and other tests, such as a spinal tap or electrical vision tests, may show "silent" damage in other places in the central nervous system. If this damage is identified, it could allow a full diagnosis of MS even after a single attack.
2. **Relapsing-remitting MS:** Symptoms come in the form of recurrent attacks with total or partial recovery. The periods

of disease inactivity between MS attacks are referred to as remission. Weeks, months, or even years may pass before another attack happens, followed again by a period of inactivity. Treatment with disease-modifying therapies can reduce the frequency of attacks or eliminate them entirely. Most people with MS are initially diagnosed with this form.

3. **Secondary-progressive MS:** Relapsing-remitting MS can gradually evolve into secondary-progressive MS. Attacks become less and less common but may still happen, and people gradually develop steady symptoms with deteriorated functioning over time. Secondary-progressive MS with attacks is called "active," whereas secondary-progressive MS without attacks is called "non-relapsing." Disease-modifying therapy for relapsing-remitting MS can delay and sometimes prevent secondary progressive MS, but the transition can happen even with treatment.
4. **Primary-progressive MS:** This course of MS is less common and has progressively worsening symptoms from the beginning, with no noticeable acute attacks. But there may be temporary or minor relief from symptoms.
5. **Radiologically isolated syndrome:** This is the rarest course of MS where a person has abnormal MRI results that look like MS, but doesn't have MS symptoms. MS symptoms (attacks or progression) may happen in the future.

Symptoms of MS

Early MS symptoms often include:

- Vision problems, such as double vision or optic neuritis (inflammation of the optic nerve), which causes pain with eye movement and vision loss

- Muscle weakness, often in the arms and legs, and muscle stiffness with painful muscle spasms
- Tingling, numbness, or pain in the arms, legs, trunk, or face
- Clumsiness, especially difficulty staying balanced when walking
- Bladder control problems
- Intermittent or constant dizziness

MS may also cause other symptoms, such as:

- Mental or physical fatigue
- Mood changes such as depression or difficulty with emotional expression or control
- Cognitive changes—including problems concentrating, multitasking, thinking, or learning, or difficulties with memory or judgment

Muscle weakness, stiffness, and spasms may be severe enough to affect walking or standing. In some cases, MS leads to partial or complete paralysis. People with MS often use a wheelchair, especially if they're untreated or have advanced disease. Many people with MS find that their symptoms are worse when they have a fever or are exposed to heat or following common infections.

Pain is rarely the first sign of MS, but pain often happens with optic neuritis and trigeminal neuralgia. Painful limb spasms and sharp pain shooting down the legs or around the abdomen can also be symptoms of MS.

Who is more likely to get multiple sclerosis?

Women are more likely to get MS than men. People of all races and ethnicities can get MS, but it's most common in White people.

Having a parent or sibling with MS increases the chances of a person getting MS, although MS itself isn't an inherited disorder. Research suggests hundreds of genes and gene variants (also called mutations) combine to create vulnerability to MS.

Some of these genes have been identified, and most are related to functions of the immune system. Some of the known genes are like those that have been identified in people with other autoimmune diseases, such as inflammatory bowel disease, celiac disease, type 1 diabetes, rheumatoid arthritis, or lupus.

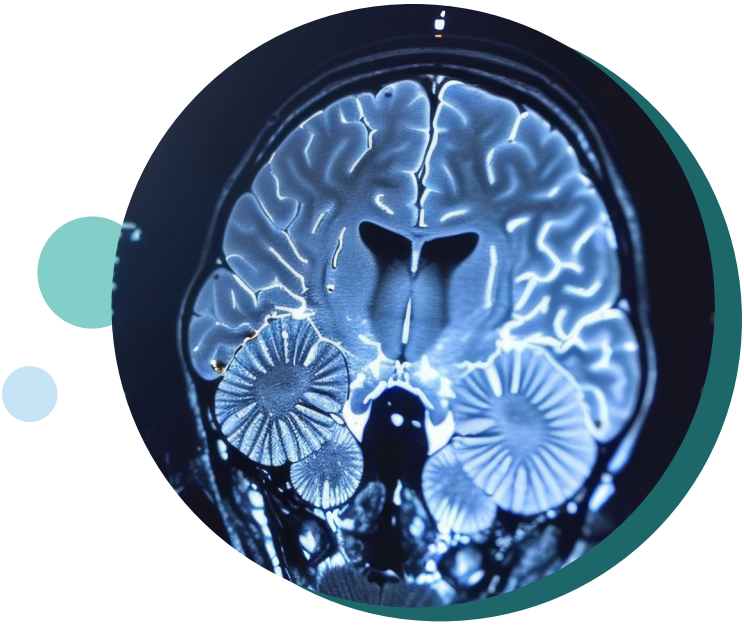
Several viruses have been found in people with MS, but the virus most consistently linked to the development of MS is the Epstein-Barr virus (EBV) which causes infectious mononucleosis. Almost everyone has been infected by EBV at some point in their lives. Only about 5% of the population hasn't been infected, and these people are at a lower risk for developing MS than those who have been infected. People who got EBV during childhood are at a lower risk of getting MS than people infected with EBV in adolescence or adulthood. But the vast majority of people who get infected with EBV aren't going to develop MS.

Research indicates people who spend more time in the sun, and those with relatively higher levels of vitamin D, are less likely to develop MS than those who don't. Also, people with MS who spend

a lot of time in the sun or have higher vitamin D levels have a less severe course of disease and fewer relapses.

Bright sunlight helps human skin produce vitamin D. Researchers believe vitamin D may help regulate the immune system in ways that reduce the risk of MS and autoimmune disorders in general. People from regions near the equator, where there is a great deal of bright sunlight, generally have a much lower risk of MS than people from temperate areas such as the U.S. and Canada, where sunshine isn't consistent throughout the year.

Studies have found people who smoke are more likely to develop MS and have a more aggressive disease course. They also tend to have more brain lesions and brain shrinkage than non-smokers.



How is multiple sclerosis diagnosed and treated?

Diagnosing MS

There's no single test to diagnose MS. Doctors use different tests to rule out or confirm the diagnosis. In addition to a complete medical history, physical examination, and a detailed neurological examination, a doctor may recommend MRI scans of the brain and spinal cord to look for the characteristic lesions of MS. A special dye or contrast agent may be injected into a vein to improve the brain images.

A doctor may also recommend:

- Lumbar puncture (sometimes called a spinal tap) to obtain a sample of cerebrospinal fluid (CSF) and examine it for proteins and inflammatory cells associated with the disease. This can also test for diseases that may look like MS.
- Evoked potential tests, which use electrodes placed on the skin and painless electric signals to measure how quickly and accurately the nervous system responds to stimulation.
- MRI of the optic nerves, optic coherence tomography (OCT), or visual evoked potentials to detect optic nerve lesions.

In most cases, doctors can diagnose MS by considering someone's symptoms and identifying characteristic MS signs on an MRI.

Learn more about neurological diagnostic tests and procedures.

Treating MS

There's no cure for MS, but there are treatments that can reduce the number and severity of relapses—and delay the long-term progression of the disease.

Doctors can prescribe corticosteroids, such as methylprednisolone, for three to five days—they're usually injected into a vein. Corticosteroids quickly and effectively suppress the immune system and reduce inflammation. They may be followed by a progressively smaller dose of oral corticosteroids (taken by mouth). Clinical trials have shown that these medicines make recovery from MS attacks faster, but they don't alter the long-term outcome of the disease.

Disease-modifying treatments

Current therapies approved by the U.S. Food and Drug Administration (FDA) for MS are designed to regulate or suppress the inflammatory reactions of the disease. They're most effective for relapsing-remitting MS or secondary-progressive MS with residual attacks. They're also effective in some cases of radiologically isolated syndrome to prevent development of clinical MS.

Infusion treatments include:

- Natalizumab (brand name: Tysabri®) works by preventing cells of the immune system from entering the central nervous system. It's very effective but is associated with an increased risk of a serious and potentially fatal viral infection of the brain called progressive multifocal leukoencephalopathy (PML). Regular

blood tests for antibodies to the virus that causes PML can help address this risk.

- Ocrelizumab (brand name: Ocrevus®) treats adults with relapsing-remitting, active secondary-progressive, or primary-progressive MS. It's currently the only FDA-approved disease-modifying therapy for primary-progressive MS. The drug targets circulating immune cells ("B cells") that have many functions, including giving rise to the cells that produce antibodies. Side effects include infusion-related reactions and increased risk of infections. Ocrelizumab may slightly increase the risk of cancer and reduce the effectiveness of some vaccines.
- Alemtuzumab (brand name: Lemtrada ®) targets proteins on the surface of immune cells. Because this drug increases the risk of autoimmune disorders, it's usually used in those who haven't responded well enough to two or more MS therapies.

Oral treatments include:

- Fingolimod (brand name: Gilenya®) reduces the MS relapse rate in adults and children. It's the first FDA-approved drug to treat MS in adolescents and children age 10 and older. The drug prevents white blood cells called lymphocytes from leaving the lymph nodes and entering the blood, brain, and spinal cord. Fingolimod may result in a slow heart rate and eye problems when first taken. Fingolimod can also increase the risk of infections, such as herpes virus infections, or in rare cases be associated with PML. Siponimod has a similar mechanism of action to fingolimod. Siponimod has been approved by the FDA to treat secondary-progressive MS.
- Dimethyl fumarate (brand name: Tecfidera®) is used to treat relapsing forms of MS. How it works isn't currently known. Side effects of dimethyl fumarate are flushing (temporary reddening

of the skin), diarrhea, nausea, and lowered white blood cell count. Diroximel fumarate (brand name: Vumerity®) is a drug similar to dimethyl fumarate, but with fewer gastrointestinal side effects.

- Teriflunomide (brand name: Aubagio®) reduces the rate of growth in the number of activated immune cells. Teriflunomide side effects can include nausea, diarrhea, liver damage, and hair loss.
- Cladribine (brand names: Mavenclad® and Leustatin® DSC) targets certain types of white blood cells that drive immune attacks in MS. The drug may increase the person's risk of developing cancer.

Injectable medicines include:

- Beta interferon drugs, which were once the most commonly used treatments for MS but are rarely used now. Potential side effects of these drugs include flu-like symptoms (which usually fade with continued therapy), depression, or elevation of liver enzymes.
- Glatiramer acetate, which can reduce the frequency of attacks in relapsing-remitting MS.

Clinical trials have shown cladribine, diroximel fumarate, and dimethyl fumarate decrease the number of relapses, delay the progression of physical disability, and slow the development of brain lesions.

Managing MS symptoms

MS causes various symptoms that can disrupt daily activities. Luckily, many of the symptoms of MS can usually be treated or managed. Neurologists (specialty doctors) with advanced training in the treatment of MS can prescribe specific medicines to treat these problems.

Eye and vision problems

Eye and vision problems are common in people with MS but rarely result in permanent blindness. Symptoms may include:

- Blurred or grayed vision
- Temporary blindness in one eye
- Loss of normal color vision
- Issues with depth perception
- Loss of vision in parts of the visual field

Uncontrolled horizontal or vertical eye movements (nystagmus), "jumping vision" (opsoclonus), and double vision (diplopia) are common in people with MS. Vision therapy exercises, special eyeglasses, and resting the eyes may be helpful.

Muscle and mobility problems

Muscle weakness and spasticity are common in MS. It's very important that people with MS stay physically active because physical inactivity can contribute to worsening stiffness, weakness, pain, fatigue, and other symptoms. Stretching and exercising muscles through water therapy, yoga, or physical therapy (PT) can help manage mild spasticity. Medicines can also help reduce spasticity.

Tremor, or uncontrollable shaking, develops in some people with MS. Assistive devices are sometimes helpful for people with tremor. Deep brain stimulation and medicines may also help.

Many people with MS have difficulty with balance and walking. The most common walking problem is ataxia—unsteady, uncoordinated movements—due to damage to the areas of the brain that coordinate muscle balance.

People with severe ataxia generally benefit from the use of a cane, walker, or other assistive device. Physical therapy (PT) can also reduce walking problems. Occupational therapy (OT) can help people learn how to walk using an assistive device or in a way that saves physical energy. The FDA has approved the drug dalfampridine to improve walking speed in people with MS.

Fatigue

Fatigue is a common symptom of MS and may be both physical (tiredness in the arms or legs) and cognitive (slowed processing speed or mental exhaustion). Daily physical activity programs of mild to moderate intensity can significantly reduce fatigue, although people should avoid excessive physical activity and minimize exposure to high temperatures.

PT and OT can sometimes help manage fatigue. PT provides personalized treatments, while OT teaches ways to use energy wisely. They also help find the right changes in the person's environment. Stress management programs or relaxation training may help some people.

Bladder control and constipation issues

Problems with bladder control and constipation may include problems with frequency of urination, urgency, or the loss of bladder control. A small number of people retain large amounts of urine. Medical treatments are available for bladder-related problems. Constipation is also common and can be treated with a high-fiber diet, laxatives, and stool softeners.

Sexual dysfunction

Sexual dysfunction can result from damage to nerves running through the spinal cord. Sexual problems may also come from MS symptoms—including fatigue, muscle symptoms, and psychological factors. Some of these problems can be corrected with medicines, and counseling (therapy) may be helpful.

Mental and emotional problems

Clinical depression is common in people with MS. MS may cause depression as part of the disease process and chemical imbalance in the brain. Depression can intensify symptoms of fatigue, pain, and sexual dysfunction. It is most often treated with cognitive behavioral therapy and selective serotonin reuptake inhibitor (SSRI) antidepressant medicines, which are less likely than other antidepressant medicines to cause fatigue.

Inappropriate and involuntary expressions of laughter, crying, or anger—called pseudobulbar symptoms—are sometimes associated with MS. But this isn't as common as in some other neurological disorders.

These expressions often don't match someone's mood. For example, people with MS may cry when they are actually happy or laugh when they aren't especially happy. The combination treatment of the medicines dextromethorphan and quinidine can treat pseudobulbar symptoms, as can other medicines such as amitriptyline or citalopram.

Cognitive problems

Cognitive impairment—a decline in the ability to think, learn, and remember—affects up to 75% of people with MS. These cognitive changes may appear at the same time as the physical symptoms, or they may develop gradually over time.

Sometimes, depression causes cognitive impairment in people with MS, and it's important to rule out depression, first. If depression does cause cognitive impairment, it can be treated. Drugs such as donepezil may be helpful in some cases.

Complementary approaches

Some people with MS report improvement in their symptoms from complementary or alternative approaches. These can include:

- Acupuncture
- Aromatherapy
- Ayurvedic medicine
- Touch and energy therapies
- Physical movement disciplines such as yoga and tai chi
- Herbal supplements
- Biofeedback

Learn more about research on complementary health approaches for MS.

Because of the risk of interactions between alternative and conventional therapies, people with MS should discuss all the therapies they are using with their doctor, especially herbal supplements. Herbal supplements have biologically active ingredients that could have harmful effects on their own or interact harmfully with other medicines.

What are the latest updates on multiple sclerosis?

The National Institutes of Health (NIH), which includes NINDS, is the leading federal funder of research on the brain and nervous system, including disorders such as MS. Other parts of NIH are funding research on topics relevant to MS—including cognitive impairment, rehabilitation strategies, and telehealth.

Although researchers haven't been able to identify the exact causes of MS, there has been excellent progress in other areas of MS research—especially in the development of new treatments to prevent what makes the disease worse. New discoveries are improving and expanding MS treatment options and helping to reduce MS-related disability.

NIH-supported research projects cover a wide range of topics. These include co-occurring conditions, mechanisms of cognitive impairment, blood-brain barrier breakdown in MS, the role of sleep and circadian rhythms, rehabilitation strategies, and telehealth. Other topics include:

- Biomarkers to accurately diagnose MS and monitor disease progression and treatment response, including blood and imaging tests
- Genetic and environmental risk factors for MS
- The role of the gut microbiome and diet in MS
- Mechanisms that underlie sex differences in the incidence and presentation of MS
- Social determinants of health that influence disease outcome and disparities in care

- The role of the immune system in MS, including its function in the central nervous system
- The role and crosstalk of various cell types in the central nervous system with relation to MS
- Basic functions of myelination, demyelination, and axonal degeneration, and strategies to overcome axonal and myelin loss

Genetic research funded by NIH is exploring the roles of "susceptibility genes"—genes that are associated with an increased risk for MS. Several candidate genes have been identified and researchers are studying their function in the nervous system to discover how they may lead to the development of MS.

Other studies aim to develop better neuroimaging tools, such as more powerful MRI methods, to diagnose MS, track disease progression, and assess treatments. Investigators are also using MRI to study the natural history of MS and to help define the mechanism of action and cause of side effects of disease modifying therapies.

Intramural research programs on MS

NIH Institutes have an active MS intramural research program among scientists working at NIH. Together, they have:

- Established and continued to develop MRI as a critical tool for examining the natural course of MS in humans, monitoring disease progression, assessing effects of treatments in clinical trials, and understanding MS biology.
- Played an important role in understanding why some people develop progressive multifocal leukoencephalopathy (PML) when taking potent MS drugs. Research teams are now developing new treatments for PML.

- Unraveled mechanisms by which viruses contribute to the development of MS.
- Conducted next-generation treatment trials targeting specific mechanisms of disease progression, using advanced MRI and fluid biomarkers as outcome measures.
- Developed the first MRI method to visualize the lymph vessels surrounding the brain, which play a critical role in neuro-immune communication.

Translational research

NIH supports translational studies to develop therapies that will stop or reverse the course of MS, focusing on pathways that modify immune system function in the peripheral and central nervous system, repair damaged myelin, or protect neurons from damage. Researchers are also developing improved disease models of MS in animals to more accurately predict drug response in human disease.

Progressive MS therapies

While scientists continue to study relapsing-remitting MS, research is also investigating treatments that slow or prevent the steady decline in function in progressive MS. In the MS-SPRINT trial, the NeuroNEXT clinical trials network tested the drug ibudilast as a potential neuroprotective drug for progressive MS. The trial showed that the drug slowed the rate of brain shrinkage as compared to a placebo. NIH intramural scientists are conducting proof-of-concept clinical trials on a key driver of clinical progression called the “chronic active lesion.”

Biomarkers for MS

Biomarkers are biological signs of disease that can be found and measured. Tests can detect biomarkers in a person's blood, through brain scans, or in other ways. They can show whether a person has a disease or condition and how it's progressing over time.

As part of a larger effort to develop and validate effective biomarkers for neurological disease, NIH is supporting two definitive multicenter MS studies:

- The Central Vein Sign in MS (CAVS-MS) study is testing whether a rapid MRI approach designed by NIH-supported scientists can use the detection of a central vein passing through brain plaques to differentiate MS from other common neurological disorders that can mimic MS. The goal is to develop a reliable imaging test for MS in order to achieve rapid yet accurate diagnosis and reduce misdiagnosis, which may affect up to 20% of people currently diagnosed with MS.
- A study to test whether a simple new blood test that measures small amounts of neuron-derived proteins (neurofilaments) can be used to predict the severity of disease and help determine whether MS drugs are working to protect brain tissues.

For more information on research about MS, check NIH RePORTER, a searchable database of current and past research projects supported by NIH and other federal agencies. RePORTER also includes links to publications and resources from these projects.

For research articles and summaries on MS, search PubMed, which contains citations from medical journals and other sites.

How can I or my loved one help to improve care for people with multiple sclerosis?

Consider participating in a clinical trial so clinicians and scientists can learn more about MS and related disorders. Clinical research with human participants helps researchers learn more about a disorder and perhaps find better ways to safely detect, treat, or prevent disease.

All types of participants are needed—those who are healthy or may have an illness or disease—of all different ages, sexes, races, and ethnicities. This helps make sure study results apply to as many people as possible and that treatments will be safe and effective for everyone who will use them.

For information about participating in clinical research visit the NINDS Clinical Trials site and NIH Clinical Research Trials and You. Learn about clinical trials currently looking for people with MS at ClinicalTrials.gov, a searchable database of current and past clinical studies and research results.

Where can I find more informaion about multiple sclerosis?

Information may be available from the following sources:

Accelerated Cure Project for Multiple Sclerosis
781-487-0008
info@acceleratedcure.org

Autoimmune Association
586-776-3900
hello@autoimmune.org

Multiple Sclerosis Association of America (MSAA)
800-532-7667
msaa@mymsaa.org

Multiple Sclerosis Foundation (MS Focus)
888-673-6287
support@msfocus.org

Myelin Repair Foundation (MRF)
info@myelinrepair.org

National Ataxia Foundation (NAF)
763-553-0020
naf@ataxia.org

National Multiple Sclerosis Society
800-344-4867
contactusnmss@nmss.org

National Organization for Rare Disorders (NORD)
203-744-0100

National Rehabilitation Information Center (NARIC)
800-346-2742

Paralyzed Veterans of America
800-555-9140
info@pva.org

Prepared by:
National Institute of Neurological Disorders and Stroke
National Institutes of Health
Department of Health and Human Services
Bethesda, Maryland 20892-2540

National Institute of
Neurological Disorders
and Stroke

Publication No. 25-NS-75
December 2025